

将下面的描述逻辑表达式转换为一阶逻辑：

一个男人的集合，这样的男人至少有三个儿子且至多有两个女儿，儿子都失业了并与医生结了婚，女儿都是物理系或数学系的教授

男人：

$$\text{Man}(x)$$

至少有 3 个儿子：

$$\exists s1 \exists s2 \exists s3 (\text{Son}(s1, x) \wedge \text{Son}(s2, x) \wedge \text{Son}(s3, x))$$

至多有 2 个女儿：

$$\neg \exists d1 \exists d2 \exists d3 (\text{Daughter}(d1, x) \wedge \text{Daughter}(d2, x) \wedge \text{Daughter}(d3, x))$$

所有儿子都失业并与医生结婚：

$$\forall s (\text{Son}(s, x) \rightarrow (\text{Unemployed}(s) \wedge \exists z (\text{Married}(s, z) \wedge \text{Doctor}(z))))$$

所有女儿都是物理系或数学系的教授：

$$\forall d (\text{Daughter}(d, x) \rightarrow (\text{Professor}(d) \wedge (\text{Physics}(d) \vee \text{Math}(d))))$$

男人至少有三个儿子且至多有两个女儿，儿子都失业了并与医生结了婚，女儿都是物理系或数学系的教授：

$$\begin{aligned} \exists x (\text{Man}(x) \wedge \exists s1 \exists s2 \exists s3 (\text{Son}(s1, x) \wedge \text{Son}(s2, x) \wedge \text{Son}(s3, x)) \wedge \neg \exists d1 \\ \exists d2 \exists d3 (\text{Daughter}(d1, x) \wedge \text{Daughter}(d2, x) \wedge \text{Daughter}(d3, x)) \wedge \forall s (\text{Son}(s, \\ x) \rightarrow (\text{Unemployed}(s) \wedge \exists z (\text{Married}(s, z) \wedge \text{Doctor}(z)))) \wedge \forall d (\text{Daughter}(d, x) \\ \rightarrow (\text{Professor}(d) \wedge (\text{Physics}(d) \vee \text{Math}(d)))))) \end{aligned}$$